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HPC remote visualization

The Helios cluster features a remote visualization solution that leverages the power of 3xNVIDIA A40 and the VirtualGL protocol.

Introduction

A specific partition has been set up on the Helios cluster to allow the execution of programs requiring resource-intensive 3D graphics rendering, possibly combined with CUDA computing.

Hardware setup:

- 3 NVIDIA Tesla A40
- 64 cores CPU
- 252 GB RAM

Specifications

- Only programs using `OpenGL` are supported
- Three user sessions are allowed at the same time
- The time limit of session is 4 hours by default
- The user can share his session with others users (see [FAQ](#))
- Connections between the client workstation and the mesocentre are not encrypted

1. Client workstation prerequisites

In order to use a graphical session on the Helios cluster, you need to install the [TurboVNC](#) software on **your workstation**.

2. Launching a graphical session

WARNING

Before using `svgl` for the first time, you must initialize your XFCE environment by starting a graphical session once with **X2Go**.

See the connection documentation:

[Access / Connect Using X2GO](#)

ON the Helios cluster, from **login node**, it is possible to request a new graphical session using the following command, when you are **in your WORK** directory:

```
$ svgl
```

TIP

- **At the first use of `svgl`, you will be asked for a password, which is different from your mesocentre password.**
- All the parameters indicated in this command will be transmitted to Slurm: it is thus possible to change Slurm parameters for the visualization job.
- It is also possible to pass arguments to the VNC server using the `VNC_OPTIONS` environment variable.

This command places the new graphical session in a Slurm queue, waiting for the requested resources to become available. The status of the session can be checked using the `squeue` command.

For example:

```
[ login@host WORK]$ squeue --me
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES
NODELIST(REASON)						
6039	visu	visu	login	R	0:27	1 m3sovisu

Once the session is running (the job goes to state R), the session connection address is displayed in the shell you're running `svg1`.

```
...
=====
First You should have installed a VNC viewer on your computer!

From your local computer you can connect to your 3D session
With vncviewer using the following adress:

mesohelios1.univ-fcomte.fr:1

If no error were displayed, the visualization server is now started.
=====

> Use CTRL+C to exit <
```

The VNC address for this session is `mesohelios1.univ-fcomte.fr:1`.

WARNING

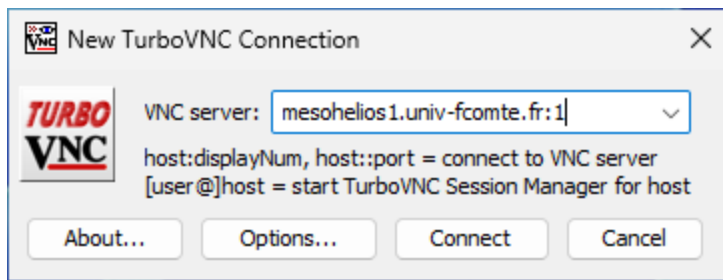
The addresses indicated change for each graphical session: it is important to refer to the one indicated at the time of execution!

3. Connection

The connection is made from your workstation, it is necessary to install TurboVNC on your workstation.

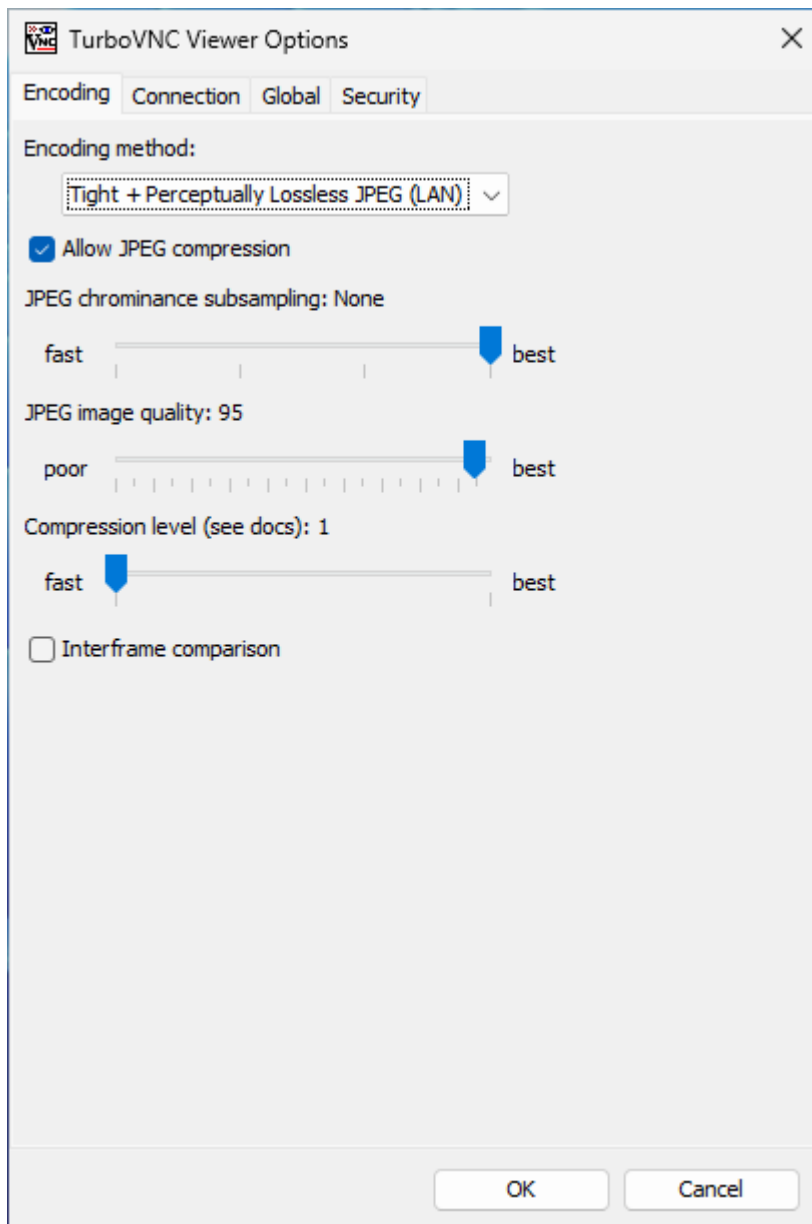
GNU Linux **Windows and Mac OS X**

On Windows or Mac systems, it is best to launch the software graphically and then use the login address when prompted.

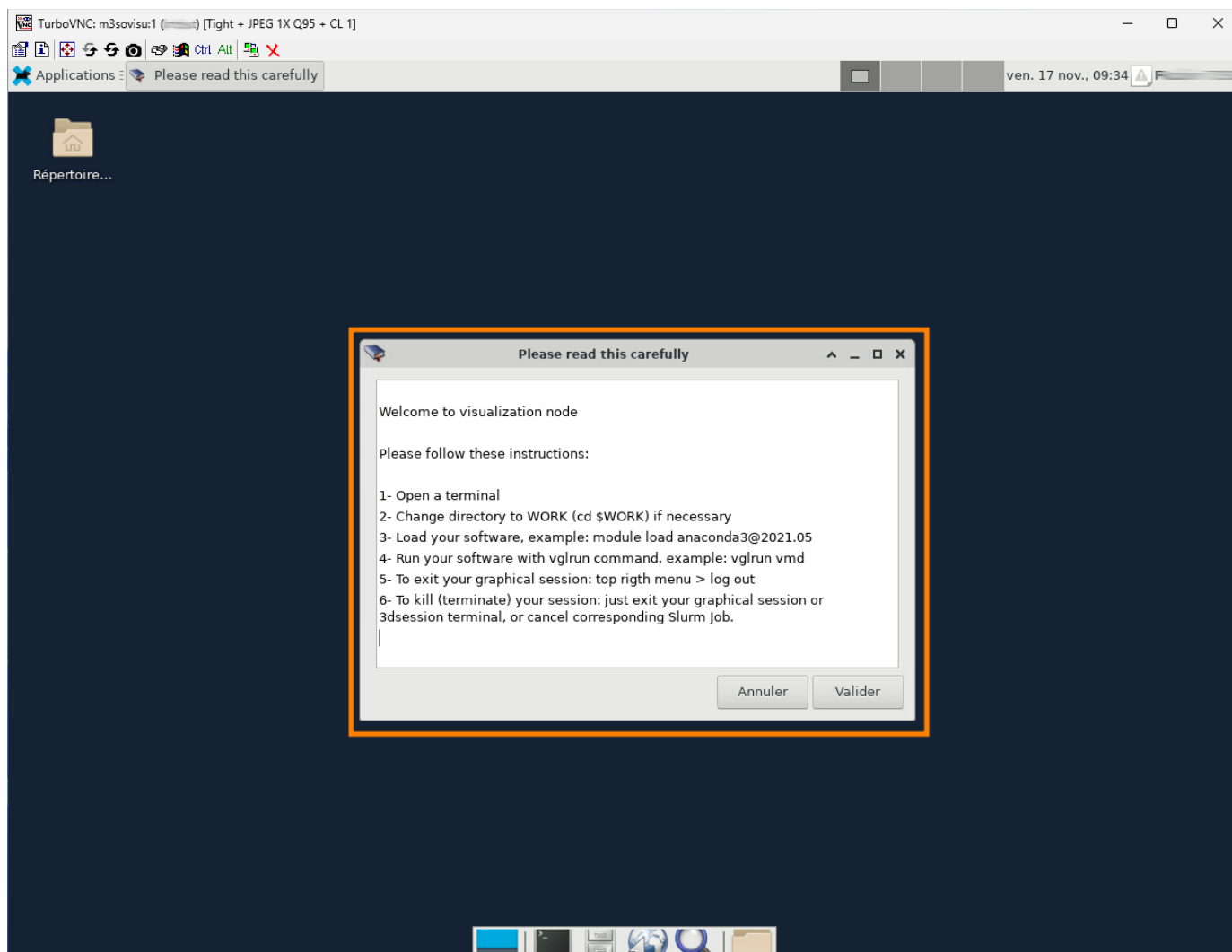


OPTIONS

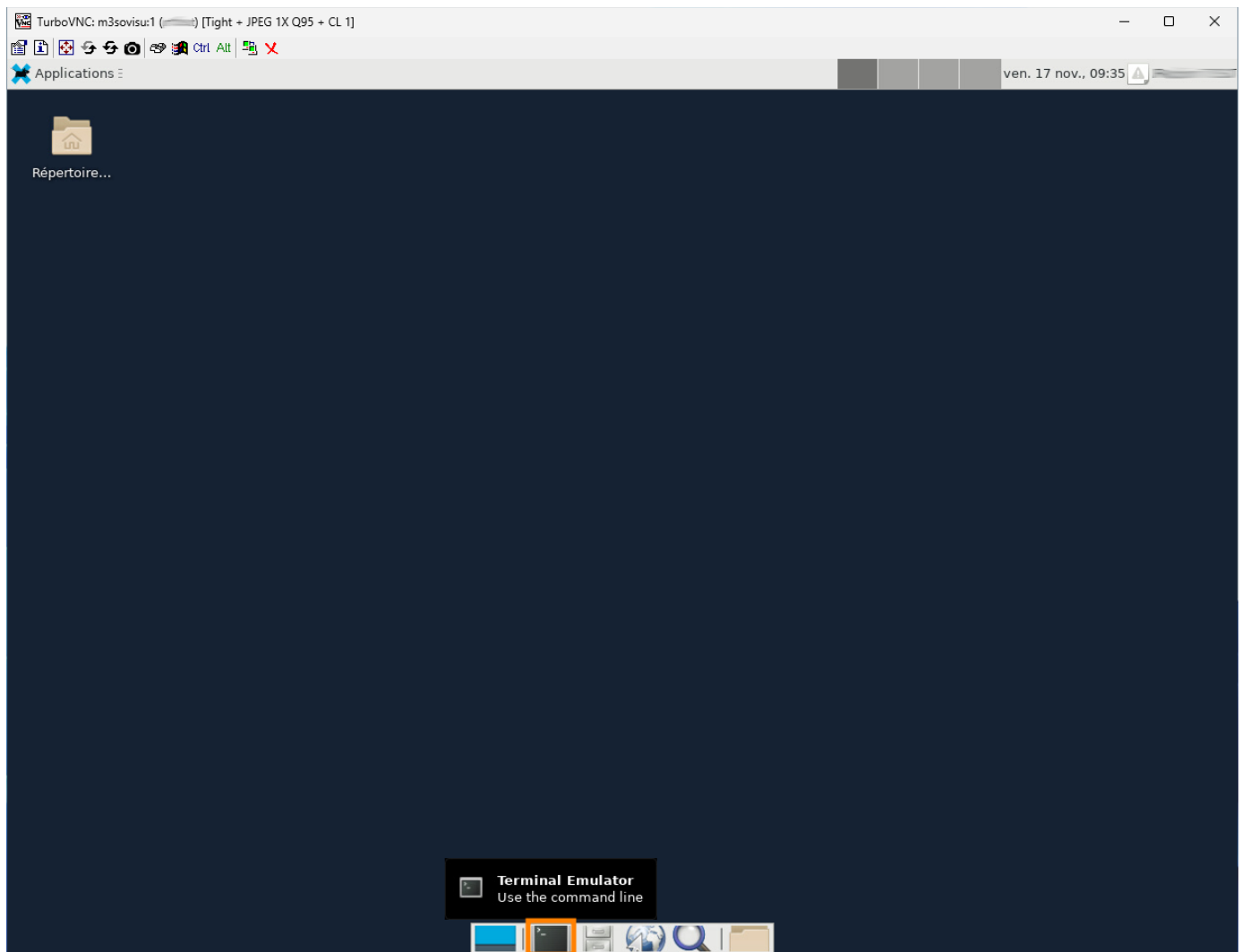
You can change the display/connection settings to suit your needs.



Once connected, a graphical session (XFCE) is initialized.



To open a terminal, one can use the **Applications** menu, or more quickly click on the terminal icon in the bottom panel.

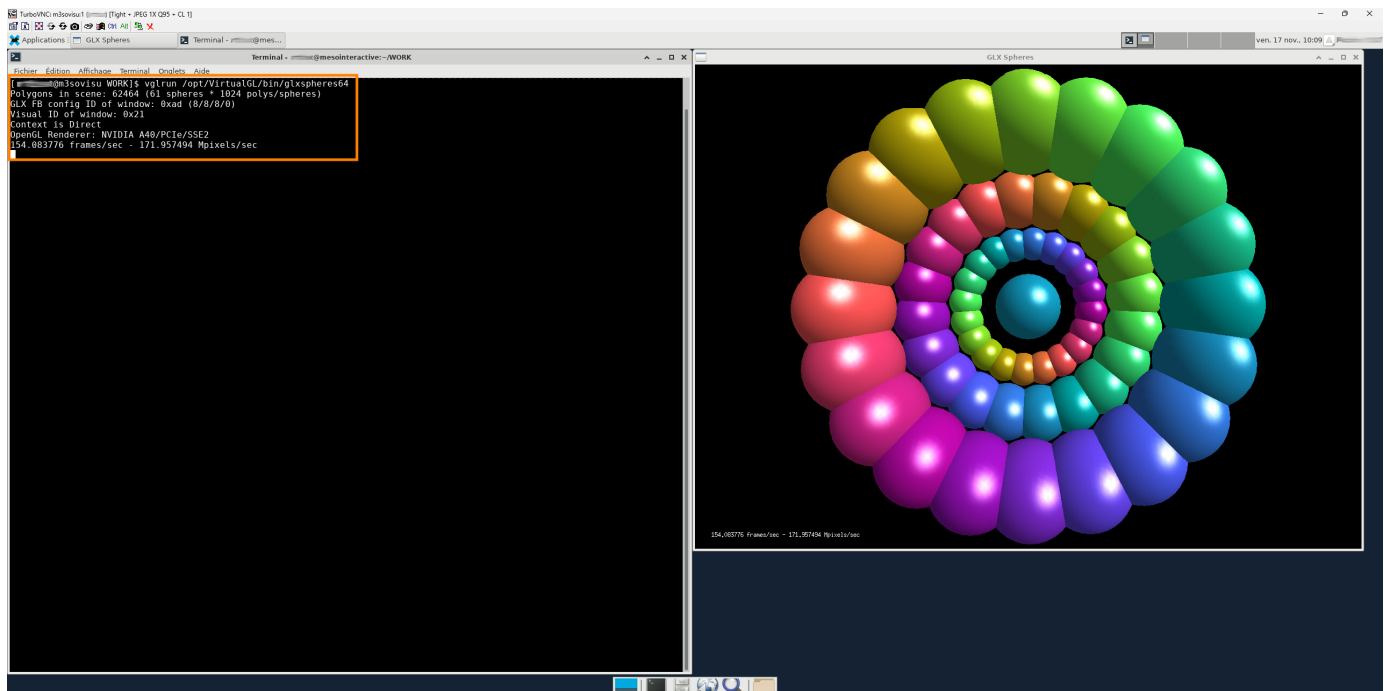


4. Running a program using 3D acceleration

In order to use the OpenGL acceleration of the system, it is necessary to prefix the program launch command with `vglrun`:

```
$ vglrun <program>
```

For example, one can see that the OpenGL renderer is a NVIDIA A40 GPU:



5. Exit the session

It is possible at any time to quit the session by going to the top right menu with your name (or **System user**) and then **Log out...** the session.

Or simply stop the Slurm job corresponding to your session, to do so you can type **CTRL + C** in the terminal where you run `svg1`, it will normally stop your job, or simply execute:

```

$ squeue --me | grep visu # to get the Slurm job id, it is the first value
$ scancel <jobid>

```

⚠ WARNING

Quitting the session will stop the Slurm job, and thus kill the programs running in the graphical session. If you want to avoid this, simply close the VNC client itself, so that you can reconnect to the graphical session later.

FAQ

- I forgot my VNC password, how can I reset it?
- How do I change the screen resolution?
- Is it possible to share screen with other users?
- My graphical session needs more than 4 hours! How do I request more time?
- Is it possible to temporarily close my session?
- How to quit/kill my session?
- How to display the `vncviewer` settings?

I forgot my VNC passowrd, how can I reset it?

You can reset your VNC password using the command `rm $WORK/.vnc/passwd` and then `svg1` will ask you for a new password.

How do I change the screen resolutions?

Before launching `svg1`, you must pass the parameter to the server via the `VNC_OPTIONS` environment variable. for example for a resolution of 1680x1050:

```
$ export VNC_OPTIONS="-geometry 1680x1050"
$ svg1
```

Is it possible to share screen with other users?

Yes, by providing your session URL and **session** password.

My graphical session needs more than 4 hours! How do I request more time?

You can request more time (for example 6 hours) with the command:

```
$ svg1 -t 6:00:00
```

Is it possible to temporarily close my session?

Yes, the graphical session is a Slurm job, just disconnect/reconnect the `vncviewer` application without exiting session.

How to quit/kill my session?

There are 2 solutions:

1. In the graphical session, click the top right menu then `Log out...`
2. On the login node of Helios, stop the Slurm job using `scancel <jobid>`

How to display the `vncviewer` settings?

By pressing the `F8` key.